

HE2003 Econometrics I: Final Revision

Wenjie Wang

**Economics, School of Social Sciences
Nanyang Technological University**

Nov 11, 2024

Revision

Note: STATA commands and outputs will NOT be tested in the final exam.

Three Categories: No, Maybe, and Yes

Week 1

1. Motivation of studying Econometrics: No
2. Correlation and causality: No
3. Randomized experiment: Maybe
4. Introduction of Test Score vs. Class Size: No

Week 2

1. Definition of the linear regression model: **Yes!**
2. Minimization problem of the OLS estimator: Maybe
3. Derivation of the OLS estimator: Maybe
4. Derivation using the OLS residuals and predicted values: Maybe

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Week 3

1. OLS Assumption 1: Maybe
2. OLS Assumption 3: No
3. Selection bias or survivorship bias: No

Week 4:

1. OLS Assumption 2: No
2. Measure of fit R^2 : **Yes!**
3. Derivations with respect to R^2 : Maybe
4. The expected value and variance of the sample average $\bar{Y}$: No

Week 5:

1. Best linear unbiased estimator and Gauss-Markov Theorem: Maybe
2. Regression with a binary variable (dummy variable): Maybe
3. Type I and Type II errors: **Yes!**

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Week 6:

1. t-test for the OLS estimator: **Yes!**
2. The large-sample distribution of the t-test under null and alternative: Maybe
3. Confidence interval and p-value: **Yes!**
4. Homoskedasticity and heteroskedasticity: **Yes!**

Week 7:

1. Omitted variable bias: Maybe
2. Definition of the multiple regression model: Maybe
3. Properties of the OLS estimators in the multiple regression model: Maybe

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Week 9:

1. Multicollinearity and dummy variable trap: **Yes!**
2. Hypothesis testing and confidence interval for a single coefficient in multiple regression: **Yes!**

Week 10:

1. Definition of joint null hypothesis and alternative hypothesis: Maybe
2. Why t-test does not work for testing joint hypothesis: No
3. Restricted and unrestricted regression model: Maybe
4. Homoskedasticity-robust F test: **Yes!**

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Week 11:

1. Definition of the chi-squared distribution: No
2. Heteroskedasticity-robust F test: No
3. Overall-regression F test: Maybe
4. Difference-in-differences (DID) and interaction regression: **Yes!**
5. Card and Krueger's study on minimum wage and employment with DID: Maybe

Week 12:

1. Interpretation of the linear-log, log-log, and log-linear models: **Yes!**
2. AJR's study of economic development using European settlers' mortality rates: Maybe